SOFE3370 – Final Project
Battery Pack SOH Prediction using Linear Regression
Group: 4

1. Overview

The project implements Linear Regression to forecast battery pack State of Health (SOH) through analysis of PulseBat Dataset voltage measurements (U1–U21).
The project uses Linear Regression to analyze voltage data (U1–U21) from the PulseBat Dataset for State of Health (SOH) prediction of used battery packs.
The goal of this project involves data cleaning and voltage aggregation, followed by regression model training and performance assessment.

2. Data Processing

The dataset contains 21 voltage features, which range from U1 to U21.
The removal of invalid data points and missing values is followed by the SOH calculation of each pack through normalized cell voltage averages.
The clean dataset received preprocessing before being exported as preprocessed_dataset.csv for training purposes.

3. Model and Results

The model implementation used LinearRegression from scikit-learn for its development.
StandardScaler scaled the data after the 80-20 split between training and testing sets.

Model performance:
$R^2 = 0.5981$
MSE = 0.002076
MAE = 0.036552

The model demonstrates strong predictive accuracy through its ability to explain 60% of SOH variations while maintaining low error rates.

4. Visualization

The generated soh_prediction_plot.png file shows Actual vs Predicted SOH values to verify the model's battery health prediction accuracy.

Conclusion

The project achieves its goal by combining data preprocessing with model training and performance assessment to predict battery SOH through Python implementation.
The results are automatically stored in the results folder for users to access.